8.3.1 A-conjugate directions ¶

8.3.1 A-conjugate directions, Part 1

fit width

Let's start our generic descent method algorithm with $x^{(0)} = 0$. Here we do not use the temporary vector $q^{(k)} = Ap^{(k)}$ so that later we can emphasize how to cast the Conjugate Gradient Method in terms of as few matrix-vector multiplication as possible (one to be exact).

$\mathbf{Given:}\ A, b$	$\mathbf{Given:}\ A, b$
$x^{(0)} := 0$	$x := 0$
$r^{(0)} := b - Ax^{(0)} (= b)$	$r := b$
$k := 0$	
$\mathbf{while}\ r^{(k)} \neq 0$	$\mathbf{while}\ r \neq 0$
$\quad p^{(k)} :=$ next direction	$\quad p :=$ next direction
$\quad \alpha_k := \frac{p^{(k)\,T} r^{(k)}}{p^{(k)\,T} A p^{(k)}}$	$\quad \alpha := \frac{p^T r}{p^T A p}$
$\quad x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$	$\quad x := x + \alpha p$
$\quad r^{(k+1)} := r^{(k)} - \alpha A p^{(k)}$	$\quad r := r - \alpha A p$
$\quad k := k + 1$	
$\mathbf{endwhile}$	$\mathbf{endwhile}$

Figure 8.3.1.1. Generic descent algorithm started with $x^{(0)} = 0$. Left: with indices. Right: without indices.

Now, since $x^{(0)} = 0$, clearly

$$x^{(k+1)} = \alpha_0 p^{(0)} + \cdots + \alpha_k p^{(k)}.$$

Thus, $x^{(k+1)} \in \mathrm{Span}(p^{(0)}, \ldots, p^{(k)})$.

It would be nice if after the kth iteration

$$f(x^{(k+1)}) = \min_{x \in \mathrm{Span}(p^{(0)}, \ldots, p^{(k)})} f(x) \tag{8.3.1}$$

and the search directions were linearly independent. Then, the resulting descent method, in exact arithmetic, is guaranteed to complete in at most n iterations, This is because then

$$\mathrm{Span}(p^{(0)}, \ldots, p^{(n-1)}) = \mathbb{R}^n$$

so that

$$f(x^{(n)}) = \min_{x \in \mathrm{Span}(p^{(0)}, \ldots, p^{(n-1)})} f(x) = \min_{x \in \mathbb{R}^n} f(x)$$

and hence $Ax^{(n)} = b$.

Unfortunately, the Method of Steepest Descent does not have this property. The next approximation to the solution, $x^{(k+1)}$ minimizes $f(x)$ where x is constrained to be on the line $x^{(k)} + \alpha p^{(k)}$. Because in each step $f(x^{(k+1)}) \leq f(x^{(k)})$, a slightly stronger result holds: It also minimizes $f(x)$ where x is constrained to be on the union of lines $x^{(j)} + \alpha p^{(j)}$, $j = 0, \ldots, k$. However, unless we pick the search directions very carefully, that is not the same as it minimizing over all vectors in $\mathrm{Span}(p^{(0)}, \ldots, p^{(k)})$.

8.3.1 A-conjugate directions, Part 2

fit width

We can write (8.3.1) more concisely: Let

$$P^{(k-1)} = \begin{pmatrix} p^{(0)} & p^{(1)} & \cdots & p^{(k-1)} \end{pmatrix}$$

be the matrix that holds the history of all search directions so far (as its columns) . Then, letting

$$a^{(k-1)} = \begin{pmatrix} \alpha_0 \\ \vdots \\ \alpha_{k-1} \end{pmatrix},$$

we notice that

$$x^{(k)} = \begin{pmatrix} p^{(0)} & \cdots & p^{(k-1)} \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \vdots \\ \alpha_{k-1} \end{pmatrix} = P^{(k-1)} a_{k-1}. \tag{8.3.2}$$

Homework 8.3.1.1. Let $p^{(k)}$ be a new search direction that is linearly independent of the columns of $P^{(k-1)}$, which themselves are linearly independent. Show that

$$\begin{array}{l} \min_{x \in \text{Span}(p^{(0)},\ldots,p^{(k-1)},p^{(k)})} f(x) = \min_y f(P^{(k)} y) \\ \quad = \min_y \left[\frac{1}{2} y_0^T P^{(k-1)\,T} A P^{(k-1)} y_0 - y_0^T P^{(k-1)\,T} b \right. \\ \quad\quad \left. + \psi_1 y_0^T P^{(k-1)^T} A p^{(k)} + \frac{1}{2} \psi_1^2 p^{(k)\,T} A p^{(k)} - \psi_1 p^{(k)\,T} b \right], \end{array}$$

where $y = \left(\frac{y_0}{\psi_1} \right) \in \mathbb{R}^{k+1}$.

▼ Hint ▼ Solution

$\min_{x \in \text{Span}(p^{(0)},\ldots,p^{(k-1)},p^{(k)})} f(x)$

$= \quad <$ equivalent formulation $>$

$\min_y f(\left(\begin{array}{c|c} P^{(k-1)} & p^{(k)} \end{array} \right) y)$

$= \quad <$ partition $y = \left(\frac{y_0}{\psi_1} \right)$ $>$

$\min_y f(\left(\begin{array}{c|c} P^{(k-1)} & p^{(k)} \end{array} \right) \left(\frac{y_0}{\psi_1} \right))$

$= \quad <$ instantiate f $>$

$$\begin{array}{l} \min_y \left[\frac{1}{2} \left[\left(\begin{array}{c|c} P^{(k-1)} & p^{(k)} \end{array} \right) \left(\frac{y_0}{\psi_1} \right) \right]^T A \left(\begin{array}{c|c} P^{(k-1)} & p^{(k)} \end{array} \right) \left(\frac{y_0}{\psi_1} \right) \right. \\ \quad \left. - \left[\left(\begin{array}{c|c} P^{(k-1)} & p^{(k)} \end{array} \right) \left(\frac{y_0}{\psi_1} \right) \right]^T b \right]. \end{array}$$

$= \quad <$ multiply out $>$

$\min_y \left[\frac{1}{2} \left[y_0^T P^{(k-1)\,T} + \psi_1 p^{(k)\,T} \right] A \left[P^{(k-1)} y_0 + \psi_1 p^{(k)} \right] - y_0^T P^{(k-1)\,T} b - \psi_1 p^{(k)\,T} b \right]$

$= \quad <$ multiply out some more $>$

$$\begin{array}{l} \min_y \left[\frac{1}{2} y_0^T P^{(k-1)\,T} A P^{(k-1)} y_0 + \psi_1 y_0^T P^{(k-1)\,T} A p^{(k)} \right. \\ \quad \left. + \frac{1}{2} \psi_1^2 p^{(k)\,T} A p^{(k)} - y_0^T P^{(k-1)^T} b - \psi_1 p^{(k)\,T} b \right] \end{array}$$

$= \quad <$ rearrange $>$

$$\begin{array}{l} \min_y \left[\frac{1}{2} y_0^T P^{(k-1)\,T} A P^{(k-1)} y_0 - y_0^T P^{(k-1)\,T} b + \psi_1 y_0^T P^{(k-1)^T} A p^{(k)} \right. \\ \quad \left. + \frac{1}{2} \psi_1^2 p^{(k)\,T} A p^{(k)} - \psi_1 p^{(k)\,T} b \right]. \end{array}$$

$$x \in \text{Span}(p^{(0)}, \ldots, p^{(k-1)}, p^{(k)})$$

if and only if there exists

$$y = \begin{pmatrix} y_0 \\ \psi_1 \end{pmatrix} \in \mathbb{R}^{k+1} \text{ such that } x = \left(\begin{array}{c|c} P^{(k-1)} & p^{(k)} \end{array} \right) \left(\frac{y_0}{\psi_1} \right).$$

8.3.1 A-conjugate directions, Part 3

fit width

Now, *if*

$$P^{(k-1)\,T} A p^{(k)} = 0$$

then

$$
\begin{array}{l}
\min_{x \in \mathrm{Span}(p^{(0)},\ldots,p^{(k-1)},p^{(k)})} f(x) \\
\quad = \quad < \text{from before} > \\
\min_y \left[\frac{1}{2} y_0^T P^{(k-1)\,T} A P^{(k-1)} y_0 - y_0^T P^{(k-1)\,T} b \right. \\
\qquad \left. + \underbrace{\psi_1 y_0^T P^{(k-1)\,T} A p^{(k)}}_{0} + \frac{1}{2} \psi_1^2 p^{(k)\,T} A p^{(k)} - \psi_1 p^{(k)\,T} b \right] \\
\quad = \quad < \text{remove zero term} > \\
\min_y \left[\frac{1}{2} y_0^T P^{(k-1)\,T} A P^{(k-1)} y_0 - y_0^T P^{(k-1)\,T} b \right. \\
\qquad\qquad\qquad \left. + \frac{1}{2} \psi_1^2 p^{(k)\,T} A p^{(k)} - \psi_1 p^{(k)\,T} b \right] \\
\quad = \quad < \text{split into two terms that can be minimized separately} > \\
\min_{y_0} \left[\frac{1}{2} y_0^T P^{(k-1)\,T} A P^{(k-1)} y_0 - y_0^T P^{(k-1)\,T} b \right] + \min_{\psi_1} \left[\frac{1}{2} \psi_1^2 p^{(k)\,T} A p^{(k)} - \psi_1 p^{(k)\,T} b \right] \\
\quad = \quad < \text{recognize first set of terms as } f(P^{(k-1)} y_0) > \\
\min_{x \in \mathrm{Span}(p^{(0)},\ldots,p^{(k-1)})} f(x) + \min_{\psi_1} \left[\frac{1}{2} \psi_1^2 p^{(k)\,T} A p^{(k)} - \psi_1 p^{(k)\,T} b \right].
\end{array}
$$

The minimizing ψ_1 is given by

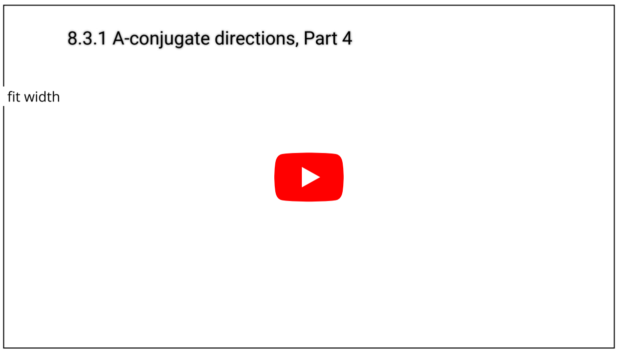
$$\psi_1 = \frac{p^{(k)\,T} b}{p^{(k)\,T} A p^{(k)}}.$$

If we pick $p^{(k)} = p^{(k)}$ and $\alpha_k = \psi_1$ then

$$x^{(k+1)} = P^{(k-1)} y_0 + \psi_1 p^{(k)} = \alpha_0 p^{(0)} + \cdots + \alpha_{k-1} p^{(k-1)} + \alpha_k p^{(k)} = x^{(k)} + \alpha_k p^{(k)}.$$

A sequence of such directions is said to be A-conjugate.

Definition 8.3.1.2. A-conjugate directions. Let A be SPD. A sequence $p^{(0)}, \ldots, p^{(k-1)} \in \mathbb{R}^n$ such that $p^{(j)\,T} A p^{(i)} = 0$ if and only if $j \neq i$ is said to be A-conjugate.



Homework 8.3.1.2. Let $A \in \mathbb{R}^{n \times n}$ be SPD.

ALWAYS/SOMETIMES/NEVER: The columns of $P \in \mathbb{R}^{n \times k}$ are A-conjugate if and only if $P^T A P = D$ where D is diagonal and has positive values on its diagonal.

▾ Answer ▾ Solution

$$
\begin{array}{l}
P^T A P \\
\quad = \quad < \text{partition } P \text{ by columns} > \\
\left(\begin{array}{c|c|c} p_0 & \cdots & p_{k-1} \end{array} \right)^T A \left(\begin{array}{c|c|c} p_0 & \cdots & p_{k-1} \end{array} \right) \\
\quad = \quad < \text{transpose} > \\
\left(\begin{array}{c} p_0^T \\ \vdots \\ p_{k-1}^T \end{array} \right) A \left(\begin{array}{c|c|c} p_0 & \cdots & p_{k-1} \end{array} \right) \\
\quad = \quad < \text{multiply out} > \\
\left(\begin{array}{c} p_0^T \\ \vdots \\ p_{k-1}^T \end{array} \right) \left(\begin{array}{c|c|c} A p_0 & \cdots & A p_{k-1} \end{array} \right) \\
\quad = \quad < \text{multiply out} > \\
\left(\begin{array}{c|c|c|c} p_0^T A p_0 & p_0^T A p_1 & \cdots & p_0^T A p_{k-1} \\ \hline p_1^T A p_0 & p_1^T A p_1 & \cdots & p_1^T A p_{k-1} \\ \hline \vdots & & \vdots & \\ \hline p_{k-1}^T A p_0 & p_{k-1}^T A p_1 & \cdots & p_{k-1}^T A p_{k-1} \end{array} \right) \\
\quad = \quad < A = A^T > \\
\left(\begin{array}{c|c|c|c} p_0^T A p_0 & p_1^T A p_0 & \cdots & p_{k-1}^T A p_0 \\ \hline p_1^T A p_0 & p_1^T A p_1 & \cdots & p_{k-1}^T A p_1 \\ \hline \vdots & & \vdots & \\ \hline p_{k-1}^T A p_0 & p_{k-1}^T A p_1 & \cdots & p_{k-1}^T A p_{k-1} \end{array} \right)
\end{array}
$$

Now, *if* the columns of P are A-conjugate, then

$$\left(\begin{array}{c|c|c|c} p_0^T A p_0 & p_1^T A p_0 & \cdots & p_{k-1}^T A p_0 \\ \hline p_1^T A p_0 & p_1^T A p_1 & \cdots & p_{k-1}^T A p_1 \\ \hline \vdots & & \vdots & \\ \hline p_{k-1}^T A p_0 & p_{k-1}^T A p_1 & \cdots & p_{k-1}^T A p_{k-1} \end{array}\right) = \quad < \text{multiply out} >$$

$$\left(\begin{array}{c|c|c|c} p_0^T A p_0 & 0 & \cdots & 0 \\ \hline 0 & p_1^T A p_1 & \cdots & 0 \\ \hline \vdots & \vdots & \ddots & \vdots \\ \hline 0 & 0 & \cdots & p_{k-1}^T A p_{k-1} \end{array}\right),$$

and hence $P^T AP$ is diagonal.

If, on the other hand, $P^T AP$ is diagonal, then the columns of P are A-conjugate.

ALWAYS

Now prove it.

Homework 8.3.1.3. Let $A \in \mathbb{R}^{n \times n}$ be SPD and the columns of $P \in \mathbb{R}^{n \times k}$ be A-conjugate.

ALWAYS/SOMETIMES/NEVER: The columns of P are linearly independent.

▾ Answer ▾ Solution

We employ a proof by contradiction. Suppose the columns of P are not linearly independent. Then there exists $y \neq 0$ such that $Py = 0$. Let $D = P^T AP$. From the last homework we know that D is diagonal and has positive diagonal elements. But then

$$\begin{array}{l} 0 \\ \quad = \quad < Py = 0 > \\ (Py)^T A (Py) \\ \quad = \quad < \text{multiply out} > \\ y^T P^T A P y \\ \quad = \quad < P^T AP = D > \\ y^T D y \\ \quad > \quad < D \text{ is SPD} > \\ 0, \end{array}$$

which is a contradiction. Hence, the columns of P are linearly independent.

ALWAYS

Now prove it!

The above observations leaves us with a descent method that picks the search directions to be A-conjugate, given in Figure 8.3.1.3.

$$\begin{array}{l} \textbf{Given:}\ A, b \\ x^{(0)} := 0 \\ r^{(0)} = b \\ k := 0 \\ \textbf{while } r^{(k)} \neq 0 \\ \quad \text{Choose } p^{(k)} \text{ such that } p^{(k)\,T} A P^{(k-1)} = 0 \text{ and } p^{(k)\,T} r^{(k)} \neq 0 \\ \quad \alpha_k := \frac{p^{(k)\,T} r^{(k)}}{p^{(k)\,T} A p^{(k)}} \\ \quad x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)} \\ \quad r^{(k+1)} := r^{(k)} - \alpha_k A p^{(k)} \\ \quad k := k+1 \\ \textbf{endwhile} \end{array}$$

Figure 8.3.1.3. Basic method that chooses the search directions to be A-conjugate.

Remark 8.3.1.4. The important observation is that if $p^{(0)}, \ldots, p^{(k)}$ are chosen to be A-conjugate, then $x^{(k+1)}$ minimizes not only

$$f(x^{(k)} + \alpha p^{(k)})$$

but also

$$\min_{x \in \mathrm{Span}(p^{(0)}, \ldots, p^{(k-1)})} f(x).$$